In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sb

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from xgboost import XGBClassifier
from sklearn import metrics

import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
pip install xgboost
```

```
Requirement already satisfied: xgboost in c:\users\malot\anaconda\lib\site-packages
(3.2.0)
Requirement already satisfied: numpy in c:\users\malot\anaconda\lib\site-packages (f
rom xgboost) (1.26.4)
Requirement already satisfied: scipy in c:\users\malot\anaconda\lib\site-packages (f
rom xgboost) (1.13.1)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
df = pd.read_csv(r'C:\Users\malot\Downloads\archive (3)\TSLA.csv')
```

In [4]:
```
df.head()
```

Out[4]:

| | Date | Open | High | Low | Close | Adj Close | Volume |
|---|---|---|---|---|---|---|---|
| **0** | 2010-06-29 | 3.800 | 5.000 | 3.508 | 4.778 | 4.778 | 93831500 |
| **1** | 2010-06-30 | 5.158 | 6.084 | 4.660 | 4.766 | 4.766 | 85935500 |
| **2** | 2010-07-01 | 5.000 | 5.184 | 4.054 | 4.392 | 4.392 | 41094000 |
| **3** | 2010-07-02 | 4.600 | 4.620 | 3.742 | 3.840 | 3.840 | 25699000 |
| **4** | 2010-07-06 | 4.000 | 4.000 | 3.166 | 3.222 | 3.222 | 34334500 |

In [5]:
```
df.shape
```

Out[5]:
```
(2956, 7)
```

In [6]:
```
df.describe()
```

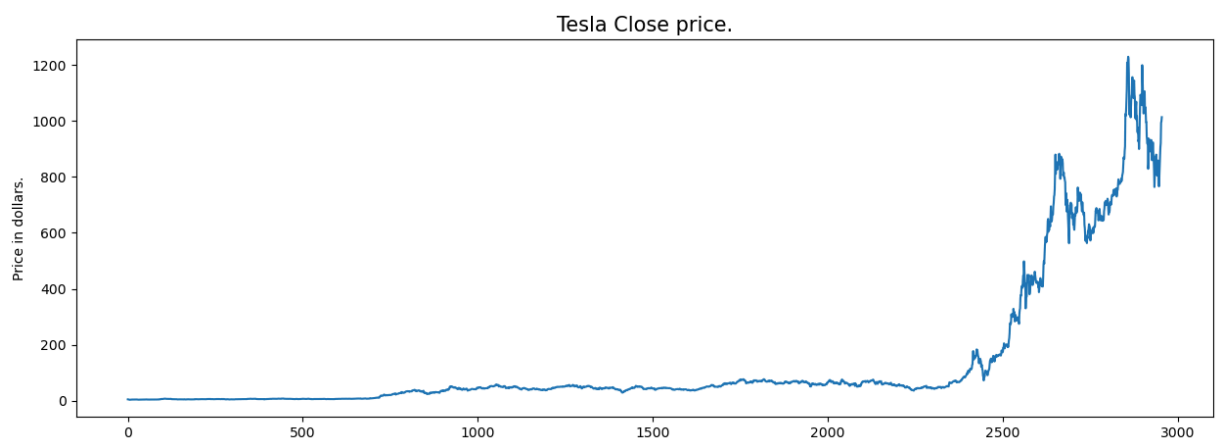
Out[6]:

| | Open | High | Low | Close | Adj Close | Volume |
|---|---|---|---|---|---|---|
| **count** | 2956.000000 | 2956.000000 | 2956.000000 | 2956.000000 | 2956.000000 | 2.956000e+03 |
| **mean** | 138.691296 | 141.771603 | 135.425953 | 138.762183 | 138.762183 | 3.131449e+07 |
| **std** | 250.044839 | 255.863239 | 243.774157 | 250.123115 | 250.123115 | 2.798383e+07 |
| **min** | 3.228000 | 3.326000 | 2.996000 | 3.160000 | 3.160000 | 5.925000e+05 |
| **25%** | 19.627000 | 20.402000 | 19.127500 | 19.615000 | 19.615000 | 1.310288e+07 |
| **50%** | 46.656999 | 47.487001 | 45.820002 | 46.545000 | 46.545000 | 2.488680e+07 |
| **75%** | 68.057001 | 69.357500 | 66.911501 | 68.103998 | 68.103998 | 3.973875e+07 |
| **max** | 1234.410034 | 1243.489990 | 1217.000000 | 1229.910034 | 1229.910034 | 3.046940e+08 |

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2956 entries, 0 to 2955
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Date       2956 non-null   object
 1   Open       2956 non-null   float64
 2   High       2956 non-null   float64
 3   Low        2956 non-null   float64
 4   Close      2956 non-null   float64
 5   Adj Close  2956 non-null   float64
 6   Volume     2956 non-null   int64
dtypes: float64(5), int64(1), object(1)
memory usage: 161.8+ KB
```

In [8]:
```
plt.figure(figsize=(15,5))
plt.plot(df['Close'])
plt.title('Tesla Close price.', fontsize=15)
plt.ylabel('Price in dollars.')
plt.show()
```




In [9]:
```
df.head()
```

Out[9]:

| | Date | Open | High | Low | Close | Adj Close | Volume |
|---|---|---|---|---|---|---|---|
| **0** | 2010-06-29 | 3.800 | 5.000 | 3.508 | 4.778 | 4.778 | 93831500 |
| **1** | 2010-06-30 | 5.158 | 6.084 | 4.660 | 4.766 | 4.766 | 85935500 |
| **2** | 2010-07-01 | 5.000 | 5.184 | 4.054 | 4.392 | 4.392 | 41094000 |
| **3** | 2010-07-02 | 4.600 | 4.620 | 3.742 | 3.840 | 3.840 | 25699000 |
| **4** | 2010-07-06 | 4.000 | 4.000 | 3.166 | 3.222 | 3.222 | 34334500 |

In [10]:
```
df[df['Close'] == df['Adj Close']].shape
```

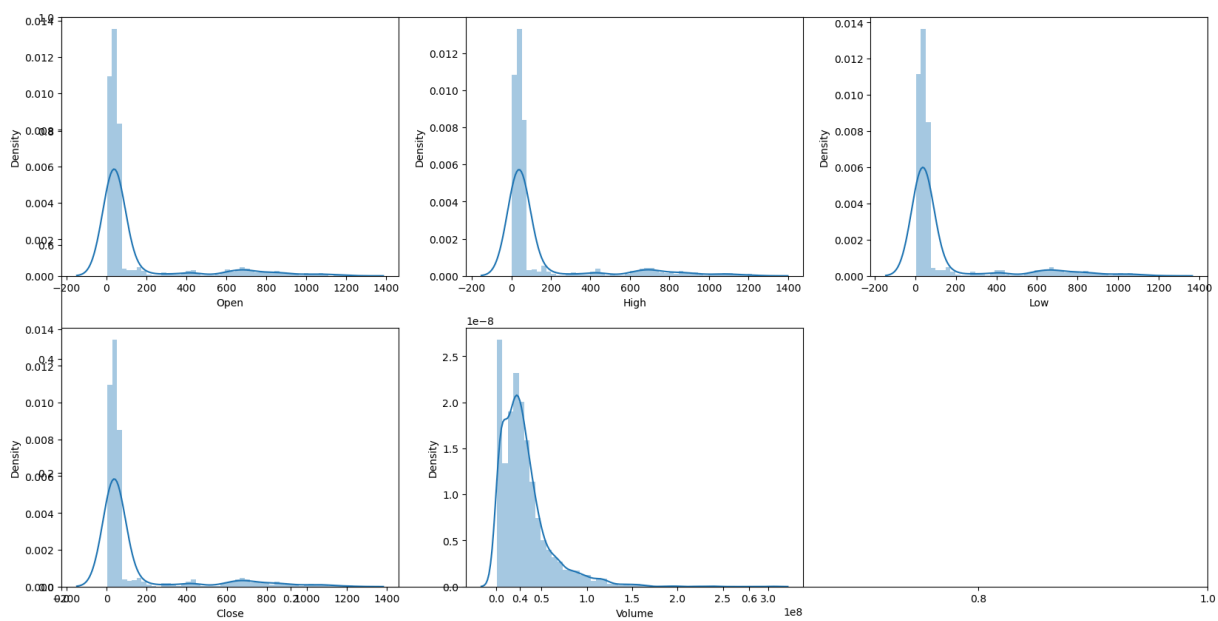
Out[10]:
```
(2956, 7)
```

In [11]:
```
df = df.drop(['Adj Close'], axis=1)
```

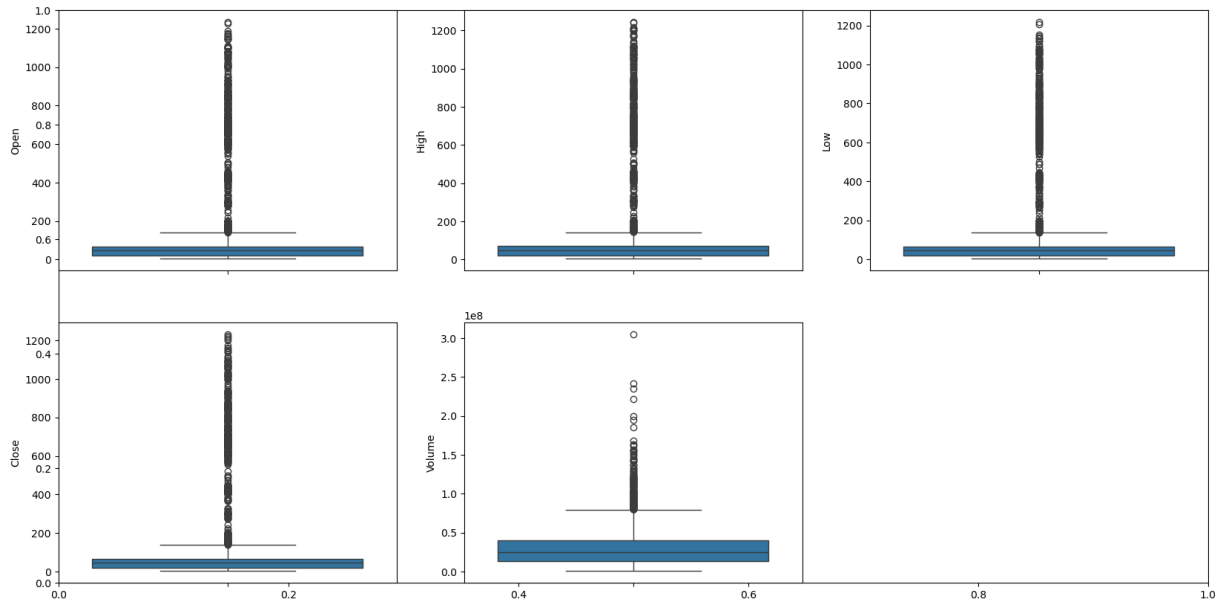
In [12]:
```
df.isnull().sum()
```

Out[12]:
```
Date      0
Open      0
High      0
Low       0
Close     0
Volume    0
dtype: int64
```

In [13]:
```
features = ['Open', 'High', 'Low', 'Close', 'Volume']

plt.subplots(figsize=(20,10))

for i, col in enumerate(features):
  plt.subplot(2,3,i+1)
  sb.distplot(df[col])
plt.show()
```



In [14]:
```
plt.subplots(figsize=(20,10))
for i, col in enumerate(features):
  plt.subplot(2,3,i+1)
  sb.boxplot(df[col])
plt.show()
```




In [15]:
```
df['Date'] = pd.to_datetime(df['Date'])

df['day'] = df['Date'].dt.day
df['month'] = df['Date'].dt.month
df['year'] = df['Date'].dt.year
```

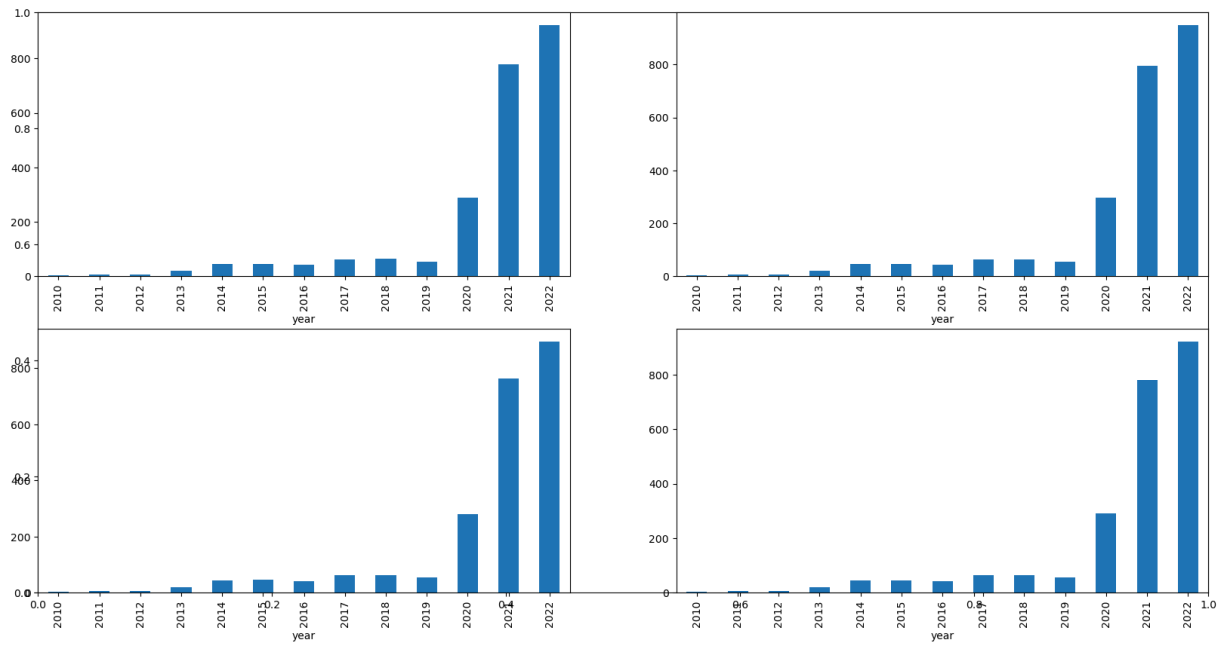
In [16]:
```
df['is_quarter_end'] = np.where(df['month']%3==0,1,0)
df.head()
```

Out[16]:

| | Date | Open | High | Low | Close | Volume | day | month | year | is_quarter_end |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2010-06-29 | 3.800 | 5.000 | 3.508 | 4.778 | 93831500 | 29 | 6 | 2010 | 1 |
| **1** | 2010-06-30 | 5.158 | 6.084 | 4.660 | 4.766 | 85935500 | 30 | 6 | 2010 | 1 |
| **2** | 2010-07-01 | 5.000 | 5.184 | 4.054 | 4.392 | 41094000 | 1 | 7 | 2010 | 0 |
| **3** | 2010-07-02 | 4.600 | 4.620 | 3.742 | 3.840 | 25699000 | 2 | 7 | 2010 | 0 |
| **4** | 2010-07-06 | 4.000 | 4.000 | 3.166 | 3.222 | 34334500 | 6 | 7 | 2010 | 0 |

In [17]:
```
data_grouped = df.drop('Date', axis=1).groupby('year').mean()
plt.subplots(figsize=(20,10))

for i, col in enumerate(['Open', 'High', 'Low', 'Close']):
```

```
  plt.subplot(2,2,i+1)
  data_grouped[col].plot.bar()
plt.show()
```



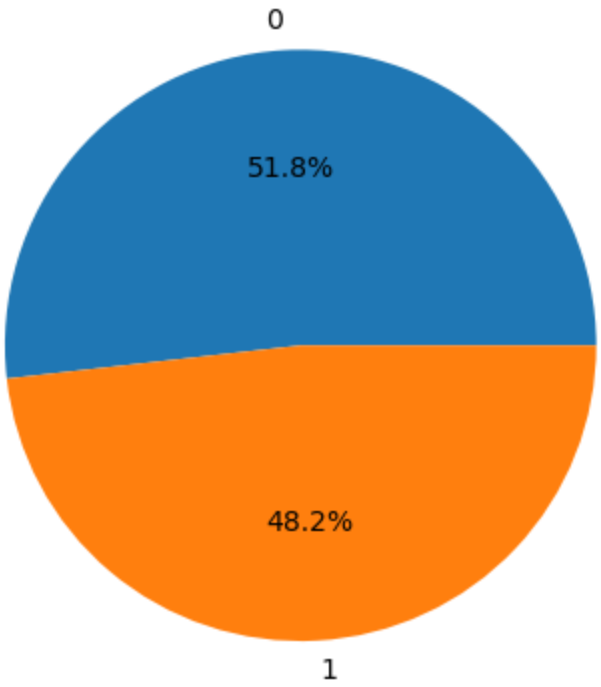

In [18]:
```
df.drop('Date', axis=1).groupby('is_quarter_end').mean()
```

Out[18]:

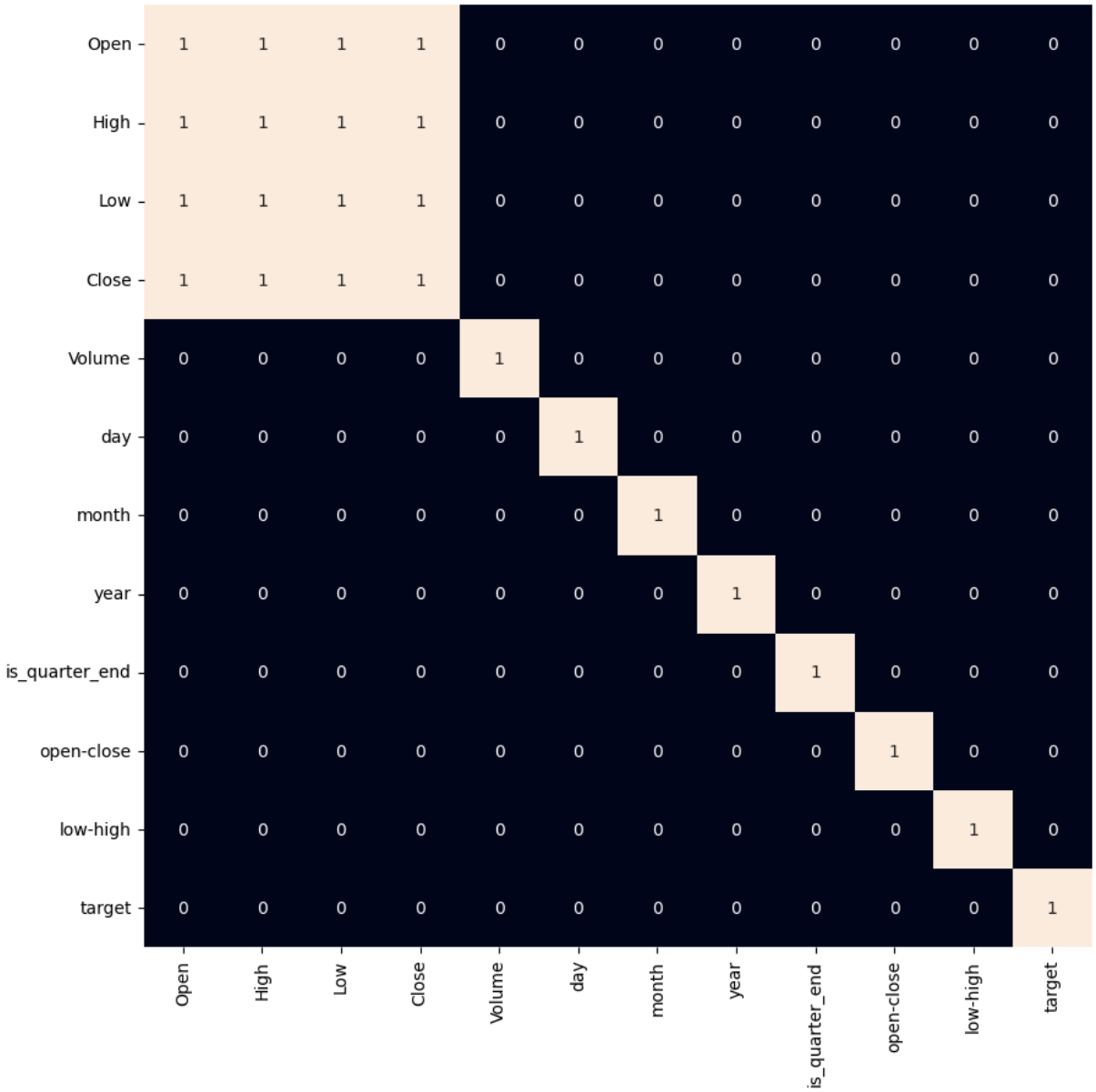
| is_quarter_end | Open | High | Low | Close | Volume | day | |
|---|---|---|---|---|---|---|---|
| **0** | 136.474690 | 139.523037 | 133.361644 | 136.531872 | 3.194378e+07 | 15.701987 | 6 |
| **1** | 143.073168 | 146.216652 | 139.506757 | 143.171146 | 3.007048e+07 | 15.736153 | 7 |

In [19]:
```
df['open-close']  = df['Open'] - df['Close']
df['low-high']  = df['Low'] - df['High']
df['target'] = np.where(df['Close'].shift(-1) > df['Close'], 1, 0)
```

In [20]:
```
plt.pie(df['target'].value_counts().values,
        labels=[0, 1], autopct='%1.1f%%')
plt.show()
```




In [21]:
```
plt.figure(figsize=(10, 10))

# As our concern is with the highly
# correlated features only so, we will visualize
# our heatmap as per that criteria only.
sb.heatmap(df.drop('Date', axis=1).corr() > 0.9, annot=True, cbar=False)
plt.show()

# This code is modified by Susobhan Akhuli
```




In [22]:
```
features = df[['open-close', 'low-high', 'is_quarter_end']]
target = df['target']

scaler = StandardScaler()
features = scaler.fit_transform(features)

X_train, X_valid, Y_train, Y_valid = train_test_split(
    features, target, test_size=0.1, random_state=2022)
print(X_train.shape, X_valid.shape)
```

```
(2660, 3) (296, 3)
```

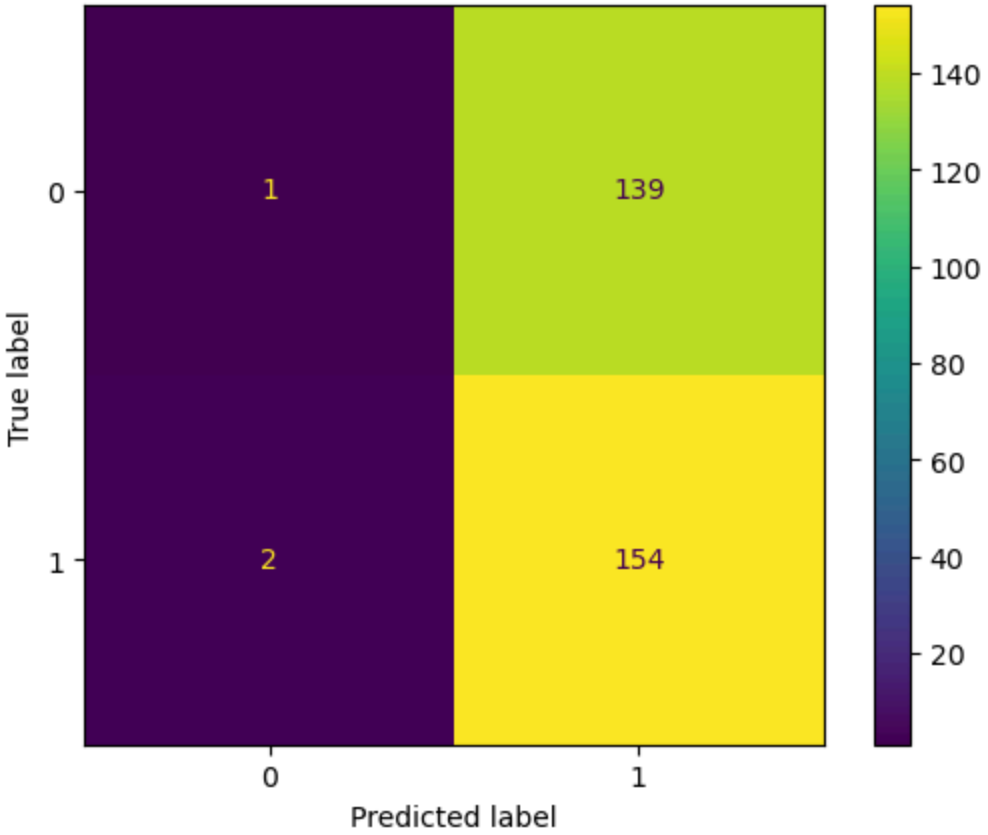
In [23]:
```
models = [LogisticRegression(), SVC(
  kernel='poly', probability=True), XGBClassifier()]

for i in range(3):
  models[i].fit(X_train, Y_train)

  print(f'{models[i]} : ')
  print('Training Accuracy : ', metrics.roc_auc_score(
```

```
    Y_train, models[i].predict_proba(X_train)[:,1]))
  print('Validation Accuracy : ', metrics.roc_auc_score(
    Y_valid, models[i].predict_proba(X_valid)[:,1]))
  print()
```

```
LogisticRegression() :
Training Accuracy :  0.5145097983728333
Validation Accuracy :  0.5385531135531135

SVC(kernel='poly', probability=True) :
Training Accuracy :  0.5003625044216484
Validation Accuracy :  0.4596611721611722

XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=None,
              n_jobs=None, num_parallel_tree=None, ...) :
Training Accuracy :  0.9240990449239477
Validation Accuracy :  0.47685439560439563
```

In [24]:

```
from sklearn.metrics import ConfusionMatrixDisplay

ConfusionMatrixDisplay.from_estimator(models[0], X_valid, Y_valid)
plt.show()

# This code is modified by Susobhan Akhuli
```


1
139
2
154
0
1
True label
0
1
Predicted label
140
120
100
80
60
40
20

In [ ]: